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3.1BSIT

Code:

```
abstract class VehicleActions {
  void move();
}

abstract class Vehicle implements VehicleActions {
  String brand;
  int speed;

  Vehicle(this.brand, this.speed);

  void displayInfo() {
    print('Brand: $brand');
    print('Speed: $speed km/h');
  }

  @override
  void move();
}

class Car extends Vehicle {
  Car(String brand, int speed) : super(brand, speed);

  @override
  void displayInfo() {
    print('Car Brand: $brand');
    print('Car Speed: $speed km/h');
    print('Number of wheels: 4');
  }

  @override
  void move() {
    print('$brand car is driving on the road.');
  }
}

class Motorcycle extends Vehicle {
  Motorcycle(String brand, int speed) : super(brand, speed);
```

```
  @override
  void displayInfo() {
    print('Motorcycle Brand: $brand');
    print('Motorcycle Speed: $speed km/h');
    print('Number of wheels: 2');
  }

  @override
  void move() {
    print('$brand motorcycle is riding on the road.');
  }
}

mixin ElectricFeature {
  void chargeBattery() {
    print('The electric vehicle is charging its battery.');
  }
}

class ElectricScooter with ElectricFeature {
  String brand;

  ElectricScooter(this.brand);

  void ride() {
    print('$brand electric scooter is ready to ride.');
  }
}

void main() {
  Car car = Car('Toyota', 120);
  Motorcycle motorcycle = Motorcycle('Honda', 90);

  print('       CAR ');
  car.displayInfo();
  car.move();

  print('\n     MOTORCYCLE ');
  motorcycle.displayInfo();
  motorcycle.move();

  print('\n     POLYMORPHISM ');
  List<Vehicle> vehicles = [car, motorcycle];
```

```
  for (Vehicle vehicle in vehicles) {
    vehicle.move();
  }

  print('\n      ELECTRIC SCOOTER ');
  ElectricScooter scooter = ElectricScooter('Xiaomi');
  scooter.ride();
  scooter.chargeBattery();
}
```

Output:

```
       CAR
Car Brand: Toyota
Car Speed: 120 km/h
Number of wheels: 4
Toyota car is driving on the road.

     MOTORCYCLE
Motorcycle Brand: Honda
Motorcycle Speed: 90 km/h
Number of wheels: 2
Honda motorcycle is riding on the road.

     POLYMORPHISM
Toyota car is driving on the road.
Honda motorcycle is riding on the road.

      ELECTRIC SCOOTER
Xiaomi electric scooter is ready to ride.
The electric vehicle is charging its battery.
```